

Ali Nasr

AI Systems Architect | R&D Engineer

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About

AI Systems Architect and R&D Engineer with extensive expertise in the design and MBSE of autonomous platforms, from agentic or intelligent decision-making to real-time robotic control.

Core Engineering Expertise

- Requirements Engineering, System & Component Architecture Design
- Prompt Engineering, Multi-Agent Orchestration, Reliability Pipelines, Human-in-the-Loop Validation
- Modeling, Identification, Training, Validation, and Simulation
- Software-in-the-loop Testing, Real-time Control Development, Signal Processing, Runtime Monitoring
- Embedded Software, Communication Systems, Sensor & Actuator Integration
- Calibration, Prototyping, and Optimization

Systems Experience

- **Generative AI:** Text Generation, Code Generation (5G Scripts & Test Cases), NL Interpretation, RAG, LLM-as-a-Judge
- **Intelligent Systems:** Autonomous Driving (Apollo Auto, End-to-End Interfuser)
- **Robotics:** Industrial, Serial & Parallel (Cable, Stewart), Humanoid, Biped, UGV, UAV (Quadcopter), Exoskeleton (Upper & Lower Limb)
- **Biological Systems:** Musculoskeletal Modeling, sEMG Analysis
- **Physical Systems:** Electrical, Hydraulic, Pneumatic, Mechanical

Skills

- **Generative AI:** LangChain, Ollama
- **Machine Learning:** ANN, RNN, CNN, RCNN, LSTM, GRU, SVM
- **Programming:** Python (NumPy, SciPy, Matplotlib, Pandas, scikit-learn), MATLAB, C++
- **System Modeling & Simulation:** MATLAB (Simulink, Simscape Multibody, Deep Learning Toolbox, Neural Networks, GA, PSO), MapleSim, LabVIEW
- **Embedded Systems:** Raspberry Pi, NVIDIA Jetson, Arduino, AVR Microcontrollers, PLCs (Siemens, Delta, Vigor)
- **Control Methods:** PID, LQR, LQG, CTM, MPC, NMPC, Feedback Linearization, Impedance, Adaptive, Sliding Mode, Fuzzy Logic, ANFIS
- **CAD & Mechanical Design:** CATIA, SolidWorks
- **Hands-on Skills:** Mechanical Fabrication, Soldering, 3D Printing

Experience

Senior R&D Engineer (GenAI System, Assurance Architect, and Design) 2023-11-01 – Present

Huawei Technologies Canada — Waterloo, Ontario, Canada

Permanent Full-time

Trustworthy Technology and Engineering Laboratory, Waterloo Complex Intelligent Systems Laboratory
LLMGuard Wrapper: A Runtime Assurance Framework for Reliable LLM Systems

- Designed and delivered a runtime monitoring and assurance architecture (GuardRailClient) for LLM-

based multi-agent systems, enabling reliable and production-ready AI behavior through integrated monitoring, validation, and correction

- Introduced a novel confidence-driven methodology that leverages token-level LogProbs aggregated into JSON-value scores to detect uncertainty and autonomously identify erroneous outputs without labeled data
- Developed a modular, scalable framework (wrapper-based integration) that seamlessly plugs into existing LLM pipelines with minimal code changes, supporting both synchronous and asynchronous execution
- Implemented an adaptive validation and recovery pipeline using LLM-as-a-Judge, multi-strategy correction (prompting, model switching, self-consistency), and optional human-in-the-loop escalation for high-risk cases
- Engineered an efficient field-level correction mechanism that selectively fixes low-confidence outputs, reducing recomputation, lowering latency, and improving system stability in real-world deployments
- Demonstrated measurable reliability improvements in a script generation system, successfully detecting ambiguous/erroneous outputs and enabling automated correction under real-world and adversarial scenarios

Uncertainty in LLMs: Theory, Methods, and Practical Reliability Frameworks

- Conducted a technical review and synthesis of uncertainty in LLMs, covering theory, state-of-the-art methods, and practical applications for improving reliability in real-world AI systems
- Analyzed and formalized key uncertainty concepts (aleatoric, epistemic, semantic, entropy-based metrics), translating research insights into actionable frameworks for inference-time risk assessment
- Evaluated and compared multiple uncertainty estimation techniques (token probability, entropy, sampling, self-check, and ensemble methods), identifying strengths, limitations, and failure modes such as "confidently wrong" predictions
- Designed and executed experiments on hallucination detection using uncertainty features, including entropy, token probabilities, and Dirichlet-based metrics, demonstrating effective model-agnostic detection without ground truth
- Developed and validated a multi-feature classification approach for hallucination detection, showing improved performance over single-metric methods and highlighting the importance of combining uncertainty signals
- Derived system-level insights for production AI systems, including the impact of temperature on uncertainty, limitations of entropy-based metrics, and the need for multi-step uncertainty tracking in agent-based workflows

Explainable & Reliable System Architecture for LLM-based Script Generator

- Architected a multi-stage LLM reliability pipeline with confidence estimation, output gating, and policy-driven decision enforcement
- Implemented probability-based and token-level uncertainty scoring to dynamically control acceptance, remediation, or escalation of LLM outputs
- Applied voting and consensus mechanisms across multiple LLM outputs to improve consistency and select highest-agreement results
- Engineered an assurance loop with temperature adjustment, prompt perturbation, and model switching for iterative refinement
- Implemented a Decision Maker using a Markov Decision Process (MDP) for policy enforcement under system constraints
- Enabled fail-safe escalation paths including re-execution with alternative models, LLM-as-Judge evaluation, and human-in-the-loop review
- Designed an explainable JSON-based output schema ensuring traceability of generated variables and intermediate decisions
- Proposed metamorphic prompt perturbation via semantic-preserving sentence swapping (Chinese inputs) to test robustness and consistency

Multi-Agent Foundation Model-based AI for Autonomous Software Testing Platform

- Designed a multi-agent system architecture leveraging foundation model-based AI (RAG-based LLMs) for a fully autonomous software testing platform
- Developed run-time accuracy and consistency monitoring mechanisms for collaborative AI agents
- Collaborated on embodied AI concepts for general-purpose intelligent assistance systems

2D Bird's-Eye View Risk Monitoring for Autonomous Driving

- Developed a CNN-based risk assessment module that processes 2D bird's-eye views of the vehicle's surroundings to predict the future driving risk
- Implemented risk detection using a 2D Time-to-Collision (TTC) algorithm for perceived object and obstructed phantom object
- Coded and integrated C++ modules into Baidu Apollo Auto ADS, enabling real-time risk evaluation and decision-making
- Verified and tested the system under diverse conditions, analyzing accident scenarios and validating robustness of the risk detection algorithms
- Animated 2D bird's-eye views and reconstructed accident cases from China for kinematic analysis and validation

Game Theory Orchestration of Multi-Agent LLM-based Reinforcement Learning

- Proposed and scoped a game-theoretic orchestration framework for multi-agent LLM-based reinforcement learning systems

Uncertainty Monitoring System for LLM-Based Generative Software

- Implemented the core uncertainty computation for the LLM outputs, forming the foundation for the monitoring system
- Developed the decision-making logic to act on uncertainty measures and partially contributed to coding the overall system
- Assisted in project planning and integration, supporting the application of uncertainty metrics to JSON-based outputs

Run-time Behavioural Validation of Self-adaptive AI-enabled Systems

- Led and coordinated the uOttawa engineering team, aligning technical efforts and ensuring milestone delivery
- Extended metamorphic testing methodologies to LLM-based software and defined practical application strategies
- Collected and formalized metamorphic relations for LLM applications to detect subtle undesirable behaviors beyond explicit failures
- Supervised development of a search-based test generation approach for autonomous driving scenarios
- Oversaw case studies implemented in CARLA simulator on the Interfuser autonomous driving system
- Directed automated offline system-level testing combining Metamorphic Testing with Evolutionary Algorithms

LLM-based System Reliability

- Guided implementation of probability extraction from LLMs within LangChain

C++ Real-Time Processing Engine for Monitoring Software

- Developed prototype of C++ modules for processing as part of the monitoring software project
- Conducted testing and validation of the C++ components to ensure reliability before integration
- Delivered ready-to-use modules to assist in integration into the larger framework

Strategic Insights and Innovations from Global Technology Conferences

- 14th International Conference on Learning Representations (ICLR), Rio de Janeiro, Brazil, Apr 2026
- 39th Annual Conference on Neural Information Processing Systems (NeurIPS), Mexico City, Mexico, Nov–Dec 2025
- 34th International Joint Conference on Artificial Intelligence (IJCAI), Montreal, Canada, Aug 2025
- 38th Annual Conference on Neural Information Processing Systems (NeurIPS), Vancouver, Canada, Dec 2024
- OpenMind Research Institute Banff Retreat, Banff, Canada, Oct 2024
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, USA, Jun 2024
- Systems Engineering Workshop for Complex Intelligent Systems, Hong Kong Science Park, Hong Kong, May 2024
- Symposium on Innovative Robotics Technologies and Applications, Hong Kong Science Park, Hong Kong, May 2024
- Strategy and Technology Workshop by Huawei R&D Teams, Shenzhen and Dongguan, China, May 2024
- 18th Annual IEEE International Systems Conference (SysCon), Montreal, Canada, Apr 2024
- 38th Annual AAAI Conference on Artificial Intelligence (AAAI), Vancouver, Canada, Feb 2024

Project Manager & Technical Interviewer Responsibilities

- Managed project workflows using Jira, tracked progress, led status updates, and prepared detailed project progress reports
- Identified team skill gaps and collaborated with leadership to define role requirements, and draft job descriptions
- Screened 1,000+ resumes, and shortlisted high-potential candidates based on technical qualifications
- Conducted 50+ remote technical interviews, evaluating problem-solving and programming ability

Senior R&D Engineer (AI Assurance in Autonomous Driving) 2023-06-01 – 2023-11-30
Huawei Technologies Canada — Waterloo, Ontario, Canada

Contract Full-time

Trustworthy Technology and Engineering Laboratory, Waterloo Complex Intelligent Systems Laboratory

Run-time Assurance (AI Consistency and Accuracy) System for Autonomous Driving Systems

- Researched failure measurement and monitoring of learning-enabled control systems across Perception, Prediction, and Planning modules
- Performed systems-theoretic process analysis and defined risk boundaries for autonomous driving systems
- Formulated and implemented C++ algorithms for monitoring, risk assessment, and secure disengagement protocols
- Designed a comprehensive run-time assurance architecture for AI consistency and accuracy validation
- Evaluated advanced autonomous driving modules by integrating Baidu Apollo Auto with CARLA Simulator using C++ and Python
- Designed pre-crash, adverse weather, and empirical simulation scenarios to validate robustness and safety

Run-time Assurance (Vehicle Safety) System for Autonomous Driving Systems

- Designed, developed, and tested vehicle recovery software for safety violations in autonomous driving systems
- Implemented proactive risk prediction and adaptive threshold tuning mechanisms
- Extracted requirements from ISO 26262, ISO 25058, and AVSC 6202103 to design compliant safety-oriented software architecture
- Developed AI system testing strategies including white-box and black-box evaluation for behavioral and activation coverage analysis

- Prototyped system components in Python and C++ and collaborated with simulation and production teams for validation

Postdoctoral Research Associate (Autonomous Control)

2023-01-01 – 2024-03-31

University of Waterloo — Waterloo, Ontario, Canada

Contract Full-time

Department of Systems Design Engineering *Real-time Embedded Controller within Different Sensors and Intelligence Control*

- Successfully designed and implemented platform real-time communication and control software on Raspberry Pi 4B (Linux environment) to support advanced application features, resulting in a 20% increase in user engagement and a 30% reduction in software crashes
- Developed machine learning-based real-time control unit with Python (multi threaded and object-oriented), resulting in improved system performance and accuracy
- Proficient in developing mid-level programs with Bluetooth, CAN-bus, and SPI connections to achieve seamless communication between hardware and software
- Conducted rigorous feature validation using hardware-in-the-loop (HIL) testing, ensuring the robustness and reliability of the system

Electrical Energy Regenerative Shoulder Exoskeletons

- Implementation of energy-regenerating actuators in robotic exoskeletons to enhance efficiency and extend battery life by converting a portion of energy typically lost during negative mechanical work into electrical energy.
- Evaluated the electrical energy regeneration capabilities of a shoulder wearable exoskeleton.
- Motor parameters were determined through experiments using a joint dynamometer system.
- Developed an electro-mechanical model for human-exoskeleton interaction focusing on energy regeneration.
- Conducted task-based experiments to measure energy regeneration during real-life usage of the exoskeleton.
- Results compared with practical usage scenarios to assess feasibility; electricity generated was relatively low due to motor characteristics, but indicated potential for energy regeneration with certain limitations.

Human3.6M+: Generated Human Motion Data to Support 3D Pose, Motion, Kinetic, and Muscle Estimation

- Developed marker-less motion capture technology for 3D human pose estimation to overcome usability constraints of traditional marker-based systems.
- Synthesized dynamics and muscle properties from existing motion capture datasets to generate training and validation data for deep learning models.
- Reconstructed motions using kinematics, dynamics, and muscle modeling while ensuring adherence to International Society of Biomechanics standards.
- Integrated musculoskeletal model with customizable anthropometric features (height, mass, athleticism level, handedness, skin temperature) to enhance realism.
- Applied methodology to Human3.6M dataset, demonstrating kinematic similarity to ground truth markers and dynamic plausibility compared to experimental data.

Postdoctoral Responsibilities

- Organized a remote international workshop with Prof. Daniel J. Rixen's group from Technical University of Munich (TUM)
- Coordinated presentations on topics ranging from humanoid robot demonstrations to advanced research in biomechanics and control systems

Myography-based Controlled 1-DoF Hand Prosthesis

- Designed an electrically controlled single degree-of-freedom (1-DoF) forearm-hand prosthesis to aid pediatric amputees after Mine Blast Injuries in Ukraine.
- Achieved precise thumb controllability through electrical component development, control system refinement, mechanical design, and final assembly.
- Prototype serves as an active pediatric prosthesis using EMG-based control rather than muscle strength, despite being limited to a single finger.
- Future improvements planned: refine 3D model, optimize electronic housing, enhance EMG control precision, and incorporate motors for full-finger control.
- Prototype is a pivotal tool for initiating prosthesis control learning in rehabilitation.

MyoBuzz: Real-time Muscle Activation Monitoring in Dynamometry Experiments

- Developed MyoBuzz, a limb-worn EMG device providing real-time feedback on muscle activation for dynamometry experiments.
- Limb-specific band with strategically positioned electrodes, visual and auditory alerts, customizable activation threshold, and adjustable sensitivity.
- Electrodes are attachable via clips; strap ensures comfort; rechargeable 3.7V battery supports one-hour operation and two-hour recharge.
- Facilitates real-time monitoring and mitigation of inadvertent muscle activation during joint torque evaluations.

Human Motion Animation Using 3Ds Max for 3D Pose Estimation Database

- Guided in setting up 3Ds Max environment and workflow for creating human motion animations for synthetic dataset generation
- Provided joint position data and guidance to support annotation for training the 3D pose estimation machine learning model

Graduate Teaching Assistant

2019-01-01 – 2023-04-30

University of Waterloo — Waterloo, Ontario, Canada

Contract Part-time

Department of Systems Design Engineering *Teaching Responsibilities*

- Delivered tutorials and undergraduate instructional support
- Explained complex engineering concepts during laboratory sessions, office hours, and via email communication
- Evaluated assignments, projects, and examinations to assess student learning outcomes

Course List **Graduate Research Assistant (BioMechatronic Systems)** 2018-09-01 – 2022-12-31

University of Waterloo — Waterloo, Ontario, Canada

Contract Full-time

Motion Research Group *Design, Dynamics, and Control of Active-Passive Exoskeletons Robotic*

- Developed and verified adjustable biomechanical 20-DoF model using MATLAB and MapleSim
- Designed, optimized, and developed actuator of wearable robot in accordance with ISB standards
- Conducted research and developed predictive models using ANN, RNN, CNN, and RCNN to estimate biomechanical (kinematic and kinetic) variables from noisy electromyography sensor data
- Developed novel assist-as-needed control regulations for wearable robots using model-based predictive control and fuzzy logic techniques
- Implemented torque and motion real-time control of active wearable robots with sEMG sensors and CAN-bus connection, leading to a 15% improvement in accuracy and a 25% increase in robot speed
- Designed and implemented safety measures for autonomous systems to ensure compliance with human testing, industry regulations, and standards, resulting in successful attainment of safety approval and ethics certificate

Active-passive Shoulder Exoskeleton

- Developed active-passive shoulder exoskeleton integrating passive mechanisms and powered actuators to optimize human movement assistance.
- Established multibody kinematics and dynamics model for upper body, implemented optimal controller for reducing muscular activation torque.
- Optimized exoskeleton design to enhance user comfort and reduce fatigue while maintaining functional assistance.
- Results demonstrate substantial reduction in fatigue and functional improvement over fully passive or fully active systems.

MuscleNET: Biomechatronic Electromyography Signal Processing

- Introduced MuscleNET, a set of ML models (ANN, RNN, CNN, RCNN) mapping sEMG signals to joint angle, velocity, acceleration, torque, and activation torque.
- RobustMuscleNET RNN predicts control-oriented signals in real-time with 92% regression accuracy for joint angle and 86.5% for assistive torque.
- InverseMuscleNET resolves redundancy in inverse muscle models using RNNs, achieving 88–91% normalized regression between estimated and experimental muscle activation.
- Enables fast, direct estimation for real-time applications in rehabilitation and sports evaluation without invasive electrodes.

Scalable Human Musculoskeletal Model

- Developed a MapleSim biomechanics-based musculoskeletal (MSK) model covering upper and lower body movements for biomechanical analysis.
- Estimated biomechanical joint torque using muscle torque generators with active and passive components.
- Created adjustable multibody kinematic model accommodating diverse anthropometric measurements (sex, age, body mass, height, dominant side, physical activity, skin temperature) with 40 DoF.
- Validated model equations through simulations of joint range of motion and torque, demonstrating robust agreement with existing research findings.

Conferences & Academic Activities

- 18th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering, Paris, France, May 2023
- Waterloo Robotics Day, Waterloo, Canada, March 2023
- 6th International Conference on Multibody System Dynamics, New Delhi, India, October 2022
- CRE-MSD Conference on Manual Handling, Mississauga, Canada, October 2022
- 5th North American Congress on Biomechanics, Ottawa, Canada, August 2022
- SYDE Graduate Symposium, Waterloo, Canada, June 2022
- 45th American Society of Biomechanics Annual Conference, Virtual, August 2021
- 45th Mechanisms and Robotics Conference, Virtual, August 2021
- 17th International Conference on Multibody Systems, Nonlinear Dynamics, and Control, Virtual, August 2021
- 7th International Conference of Control, Dynamic Systems, and Robotics, Virtual, November 2020
- 16th Annual Ontario Biomechanics Conference, Nottawasaga, Canada, March 2019

R&D Mid-Level Engineer (Robotics)

2016-01-01 – 2018-08-31

Techno Niro — Dorcheh, Esfahan, Iran

Contract Full-time Spatial 6 DoF 8 Cable-Driven Parallel Robot

- Conducted comprehensive inverse kinematic and dynamic modeling and analysis for the Spatial 6 DoF 8 Cable-Driven Parallel Robot.
- Validated the developed models using the SimMechanics Toolbox.
- Designed an optimal cable structure attachment for the robot.

- Simulated PD Partial, LQR, Robust, and Adaptive Control in the presence of noise and disturbance.

3 DoF Wrist Rehabilitation Robot

- Designed conceptual and mechanical parts for the 3 DoF Wrist Rehabilitation Robot and conducted kinematic and dynamic modeling.
- Implemented modified impedance control for enhanced human-robot interaction.
- Prototyped the wrist rehabilitation robot (Jan 2016 – Jan 2017), integrating modeling, mechanical design, and control for testing.

Graduate Marker

2015-09-01 – 2015-11-30

K. N. Toosi University of Technology — Tehran, Tehran, Iran

Remote

Faculty of Mechanical Engineering *Teaching Responsibilities*

- Assessed student learning through evaluating assignments, projects, and exams

Course List **Graduate Research Assistant (System Control)**

2014-09-01 – 2015-12-31

K. N. Toosi University of Technology — Tehran, Tehran, Iran

Contract Part-time

Advanced Robotics & Automated Systems (ARAS) Laboratory *Multiple Impedance Control of Hybrid Parallel-Serial Manipulator for Object Manipulation*

- Designed and developed a MATLAB/Simulink model for an 8 cable-driven parallel robot prototype with a serial manipulator, and verified its performance
- Conducted a comprehensive analysis of stiffness, sensitivity, singularity, and workspace volume to optimize the structure of a constraint and suspended parallel robot, using Genetic Algorithm (GA)
- Simulated and evaluated multiple impedance control algorithms, including serial and parallel, for object manipulation and human-robot force/moment interaction
- Prototyped the 8 cable constraint parallel robot, integrated image processing (vision) and IMU modules for precise position and orientation feedback, proposed an algorithm to address sensor shortage, and successfully implemented real-time motion control

Optimal Redundancy Resolution Scheme in a Cable-Driven Parallel Robot

- Developed an enhanced kinematic modeling of pulleys and cables for more accurate representation of cable-driven parallel robots
- Improved tension distribution analysis by integrating precise cable attachment modeling into the system framework
- Implemented a basic optimization approach using fmincon for estimating actuator force distribution

Sensory Feedback and Experimental Evaluation for Cable Robots with Uncertain Jacobian

- Led the mathematical modeling, algorithm coding, and manufacturing of the prototype of the cable-driven parallel robot
- Integrated the experimental and theoretical foundations on the robot's design and optimization approach
- Assisted the team with hardware, software, and modeling efforts

Visual Servoing in a Cable Robot Using Microsoft Kinect V2 Sensor

- Developed the kinematic model and control code for the parallel cable robot
- Supported integration between the robot and the visual tracking system
- Guided the use and modification of the Microsoft Kinect sensor for experiments

Digital Image Processing of Ball-and-Plate System Using Artificial Neural Network

- Integrated camera with MATLAB for real-time ball position tracking using classical image processing

techniques

- Contributed to development of a vision-based closed-loop control system
- Improved stability and responsiveness of ball positioning on a two-axis tilting platform

R&D Junior Engineer (Robotics)

2013-09-01 – 2015-12-31

Techno Niro — Dorcheh, Esfahan, Iran

Contract Part-time 6 DoF Industrial Serial Manipulator Modeling & Control

- Conducted comprehensive kinematic and dynamic modeling and analysis for the 6 DoF Industrial Serial Manipulator, assessing workspace, dexterity, and performance
- Utilized CATIA software for the design of mechanical parts, performed finite element analysis, and supervised the manufacturing process using CNC machines
- Prepared the hardware-in-loop of the control system using LabVIEW and MATLAB, and conducted motor parameters identification
- Implemented PID control, adjusted control coefficients for different maneuvers, and planned trajectory using ARM-controller programming

R&D Intern (Automation)

2013-06-01 – 2013-08-31

Kian Control Apadana Co. (Festo Authorized Distributor) — Isfahan, Esfahan, Iran

Internship Automation and Mechatronic System Development

- Developed the mechatronic scheme for a steel plant using ePLAN software, including designing the sensors, actuators, and wiring for the system
- Conducted extensive research on the hydraulic variable displacement pump action, exploring various modes of operation and optimizing their efficiency

Undergraduate Marker

2013-01-01 – 2013-05-31

Isfahan University of Technology — Isfahan, Esfahan, Iran

Hybrid Teaching Responsibilities

- Assessed student learning by marking assignments

Course List Undergraduate Research Assistant

2012-09-01 – 2013-08-31

Isfahan University of Technology — Isfahan, Esfahan, Iran

Contract Part-time

Department of Mechanical Engineering *Joint Space Energy Optimization of Tethered Space Robot for End-Effector Optimal Path*

- Conducted research on tethered satellite systems and tether-connected space manipulators for applications such as satellite repair and space debris removal.
- Modeled a tethered space manipulator system using a straight, rigid rod for the tether, restricting motion to the orbital plane with the platform following a circular orbit around Earth.
- Analyzed system dynamics with the end-effector tracing prescribed trajectories, monitoring tether tension to prevent compressive forces.
- Investigated feasible end-effector trajectories under torque restrictions, ensuring torques on tether attachment points were zero.
- Proposed a method to determine possible end-effector paths between two points while satisfying the zero-torque constraint on the tether.
- Developed and simulated a nonlinear control algorithm for vibration reduction in a redundant robot by adjusting cable length, validated through simulations and experiments.
- Performed comprehensive modeling of forward kinematics, workspace, and dynamics of a 4-DoF tethered space robot to evaluate performance and energy efficiency.

R&D Intern

2009-06-01 – 2013-08-31

Techno Niro — Dorcheh, Esfahan, Iran

Contract Part-time Plant Mechatronic Design, Maintenance, and Project Management

- Developed and supervised the implementation of a comprehensive mechatronic scheme for the plant, utilizing ePlan software and programming Siemens PLC for optimal performance
- Provided proactive maintenance and effective troubleshooting for mechanical systems and equipment
- Actively participated in project planning and execution, fostering collaboration to meet project deadlines
- Ensured adherence to quality standards, conducted regular inspections, and maintained accurate documentation of all mechanical systems
- Stayed abreast of industry trends and advancements, and pursued continuous learning through workshops and training

Education

Doctor of Philosophy (Ph.D.) in Systems Design Engineering 2018-09-01 – 2022-12-30
University of Waterloo — Waterloo, Ontario, Canada

- Major: Intelligent & Automated Systems; Biomechatronics Systems
- GPA: 88/100
- Thesis: [Design, Dynamics, and Control of Active-passive Upper-limb Exoskeleton Robots](#)

Master of Science (M.Sc.) in Mechatronics Engineering 2013-09-01 – 2015-12-30
K. N. Toosi University of Technology — Tehran, Tehran, Iran

- Major: Design Robotic and Mechatronic Systems
- GPA: 19.2/20
- Thesis: [Multiple Impedance Control of Hybrid Parallel-Serial Manipulator for Object Manipulation](#)

Bachelor of Science (B.Sc.) in Mechanical Engineering 2009-09-01 – 2013-08-30
Isfahan University of Technology — Isfahan, Esfahan, Iran

- Major: Mechatronics and Dynamic Systems
- GPA: 16.67/20
- Thesis: [Optimizing Joint Space Energy of Redundant Tethered Space Robots](#)

Publications

1. Yousefizadeh, Hossein, Gu, Shenghui, Briand, Lionel C., **Nasr, Ali**. *Constrained co-evolutionary metamorphic differential testing for autonomous systems with an interpretability approach*. Transactions on Software Engineering and Methodology (TOSEM) (2026).
2. Yousefizadeh, Hossein, Gu, Shenghui, Briand, Lionel C., **Nasr, Ali**. *Using cooperative co-evolutionary search to generate metamorphic test cases for autonomous driving systems*. (2026).
3. Yousefizadeh, Hossein, Gu, Shenghui, Briand, Lionel C., **Nasr, Ali**. *Using cooperative co-evolutionary search to generate metamorphic test cases for autonomous driving systems*. Transactions on Software Engineering (2025). [doi:10.1109/TSE.2025.3570897](https://doi.org/10.1109/TSE.2025.3570897)
4. **Nasr, Ali**, McPhee, John. *Harnessing biomechanical energy using shoulder exoskeleton: Electrical energy regeneration*. Journal of Mechanisms and Robotics (2025). [doi:10.1115/1.4068430](https://doi.org/10.1115/1.4068430)
5. **Nasr, Ali**, Inkol, Keaton A., McPhee, John. *Safety in wearable robotic exoskeletons: Design, control, and testing guidelines*. Journal of Mechanisms and Robotics (2025). [doi:10.1115/1.4066900](https://doi.org/10.1115/1.4066900)
6. **Nasr, Ali**, Risacher, Frederic. *Method and a system for run-time operational monitoring of an autonomous vehicle*. U.S. Patent (2024).
7. **Nasr, Ali**, Zhu, Kevin, McPhee, John. *Using musculoskeletal models to generate physically-consistent data for 3D human pose, kinematic, dynamic, and muscle estimation*. Journal of

Multibody System Dynamics (2024). [doi:10.1007/s11044-024-10021-5](https://doi.org/10.1007/s11044-024-10021-5)

8. Zhu, Kevin, **Nasr, Ali**, Wong, Alexander, McPhee, John. *3D human pose and torque estimation from monocular video*. (2024).
9. **Nasr, Ali**, McPhee, John. *Inverse kinematic, dynamic, and muscle simulation from video files: Toward markerless 3D human pose, torque, and muscle estimation*. (2024).
10. Korovkina, Vasylysa, **Nasr, Ali**, McPhee, John. *Optimal EMG placement for hand prosthetic control*. (2024).
11. Bell, Sydney, **Nasr, Ali**, McPhee, John. *General muscle torque generator model for a two degree-of-freedom shoulder joint*. Journal of Biomechanical Engineering (2024). [doi:10.1115/1.4065044](https://doi.org/10.1115/1.4065044)
12. **Nasr, Ali**, McPhee, John. *Scalable musculoskeletal model for dynamic simulations of lower body movement*. Journal of Computer Methods in Biomechanics and Biomedical Engineering (2024). [doi:10.1080/10255842.2024.2316240](https://doi.org/10.1080/10255842.2024.2316240)
13. **Nasr, Ali**, Dickerson, Clark R., McPhee, John. *Experimental study of fully passive, fully active, and active-passive upper-limb exoskeleton efficiency: An assessment of lifting tasks*. Journal of Sensors (2023). [doi:10.3390/s24010063](https://doi.org/10.3390/s24010063)
14. **Nasr, Ali**, Bell, Sydney, Whittaker, Rachel L., Dickerson, Clark R., McPhee, John. *Robust machine learning mapping of sEMG signals to future actuator commands in biomechatronic devices*. Journal of Bionic Engineering (2023). [doi:10.1007/s42235-023-00453-8](https://doi.org/10.1007/s42235-023-00453-8)
15. Haraguchi, Naoto, **Nasr, Ali**, Inkol, Keaton A., Hase, Kazunori, McPhee, John. *Human and passive lower-limb exoskeleton interaction analysis: Computational study with dynamics simulation using nonlinear model predictive control*. IEEE (2023). [doi:10.23919/SICE59929.2023.10354196](https://doi.org/10.23919/SICE59929.2023.10354196)
16. McPhee, John, **Nasr, Ali**. *Multibody system dynamics: A fundamental tool for biomechatronic system design*. (2023).
17. **Nasr, Ali**, Hashemi, Arash, McPhee, John. *Scalable musculoskeletal model for dynamic simulations of upper body movement*. Journal of Computer Methods in Biomechanics and Biomedical Engineering (2023). [doi:10.1080/10255842.2023.2184747](https://doi.org/10.1080/10255842.2023.2184747)
18. **Nasr, Ali**, Hunter, Jason, Dickerson, Clark R., McPhee, John. *Evaluation of a machine-learning-driven active-passive upper-limb exoskeleton robot: Experimental human-in-the-loop study*. Journal of Wearable Technologies (2023). [doi:10.1017/wtc.2023.9](https://doi.org/10.1017/wtc.2023.9)
19. **Nasr, Ali**, McPhee, John. *Computational evaluation of active-passive exoskeleton controllers with a scalable biomechatronics model*. (2023).
20. **Nasr, Ali**. *Design, Dynamics, and Control of Active-passive Upper-limb Exoskeleton Robots*. University of Waterloo (2022).
21. **Nasr, Ali**, Bell, Sydney, McPhee, John. *Optimal design of active-passive shoulder exoskeletons: A computational modeling of human-robot interaction*. Journal of Multibody System Dynamics (2022). [doi:10.1007/s11044-022-09855-8](https://doi.org/10.1007/s11044-022-09855-8)
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28. **Nasr, Ali**, Bell, Sydney, He, Jiayuan, Whittaker, Rachel L., Dickerson, Clark R., Jiang, Ning, McPhee, John. *MuscleNET: Mapping electromyography to kinematic and dynamic biomechanical*

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 34. **Nasr, Ali**, Arami, Arash, McPhee, John. *Optimal cost function for predicting upper-limb movement with external load*. (2019).
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 40. **Nasr, Ali**, Moosavian, S Ali A. *Multi-criteria design of 6-dof fully-constrained cable driven redundant parallel manipulator*. IEEE (2015). [doi:10.1109/ICRoM.2015.7367591](https://doi.org/10.1109/ICRoM.2015.7367591)
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 43. **Nasr, Ali**, Asadi, Farhad, Azami, Kian, Ziaei-Rad, Saeed. *Optimization of trunk angle in biped robots in generating a stable path in up-stair walking*. Iranian Society of Mechanical Engineers (ISME) (2015).
 44. **Nasr, Ali**, Shahriari, Yasser, Sadati, Sayed Hussein. *Optimization in joint space of six-legged walking robots*. Iranian Society of Mechanical Engineers (ISME) (2014).
 45. **Nasr, Ali**. *Optimizing Joint Space Energy of Redundant Tethered Space Robots*. Isfahan University of Technology (2013).

Credentials

Selected Certifications

- 2026-05-29 [Develop Your Skills with Large Language Models](#) — LinkedIn Learning
- 2026-05-29 [Introduction to Prompt Engineering for Generative AI](#) — LinkedIn Learning
- 2026-05-29 [GPT Foundations: Building AI-Powered Apps](#) — LinkedIn Learning
- 2026-03-09 [AI Text Summarization with Hugging Face](#) — LinkedIn Learning
- 2026-02-25 [Practice It: Python Data Structures](#) — LinkedIn Learning
- 2026-02-12 [Introduction to LLM Vulnerabilities](#) — LinkedIn Learning
- 2026-02-11 [Systems Engineering with SysML](#) — LinkedIn Learning

- 2026-02-06 [Artificial Intelligence Foundations: Thinking Machines](#) — LinkedIn Learning
- 2025-12-17 [Data-Driven Decision-Making for the Real World](#) — National Association of State Boards of Accountancy (NASBA)
- 2025-11-11 [Introduction to AI Governance](#) — National Association of State Boards of Accountancy (NASBA)
- 2025-11-10 [Understanding and Implementing the NIST AI Risk Management Framework \(RMF\)](#) — LinkedIn Learning
- 2025-10-08 [What Is Generative AI?](#) — National Association of State Boards of Accountancy (NASBA)
- 2025-10-03 [Compliance and Regulations for Generative AI](#) — LinkedIn Learning
- 2025-10-01 Project Manager for Technical Cooperation in Universities and Research Institutions — Huawei Canada
- 2025-09-16 [Building Secure and Trustworthy LLMs Using NVIDIA Guardrails](#) — LinkedIn Learning
- 2025-08-13 [GraphRAG Essential Training](#) — LinkedIn Learning
- 2025-08-01 [Agentic AI Fundamentals: Architectures, Frameworks, and Applications](#) — LinkedIn Learning
- 2025-07-31 [AI Trust and Safety: Navigating the New Frontier](#) — LinkedIn Learning
- 2025-07-31 [Build AI Agents and Automate Workflows with n8n](#) — LinkedIn Learning
- 2025-07-31 [Building Agents with Vertex AI](#) — LinkedIn Learning
 - ... and 122 more — see [website](#)

Awards & Honours

- 2026-06-01 1st Place in Project Management Competition on University Contracts and Research Proposals — Huawei Technologies Canada Co., Ltd.
- 2026-03-05 Innovation Excellence for Trustworthy AI Architecture in Cloud Core Networks — Huawei Technologies Canada Co., Ltd.
- 2025-12-01 2025 Future Star — Huawei Technologies Canada Co., Ltd.
- 2025-06-01 Multi-Agent AI Architecture Innovation Award — Huawei Technologies Canada Co., Ltd.
- 2025-02-01 Presentation Award at the Waterloo Innovation Forum – Staff Insights — Huawei Technologies Canada Co., Ltd.
- 2022-06-01 1st Place Award at the 2022 SYDE Graduate Student Symposium — Department of Systems Design Engineering, University of Waterloo
- 2021-09-01 3rd Place Award at Waterloo.AI GRADflix Competition — Waterloo Artificial Intelligence Institute, University of Waterloo
- 2021-08-01 Best Paper Award in the Area of Multibody Dynamics — American Society of Mechanical Engineers (ASME)
- 2020-11-01 Best Paper Award in the Fields of Control, Dynamic Systems, and Robotics — 7th International Conference of Control, Dynamic Systems, and Robotics
- 2018-09-01 \$119K Graduate Research Studentship (GRS) from the NSERC and CRC Programs (Four Years) — University of Waterloo
- 2018-09-01 \$58K International Doctoral Student Award (IDSA) — University of Waterloo
- 2016-01-01 Ranked 1st Among Mechatronics Students in the Department of Electrical Engineering — K.N. Toosi University of Technology
- 2015-11-01 2nd Place and Gold Coin Award at the Creativity Exhibition — The 3rd International Conference on Robotics and Mechatronics
- 2013-09-01 Full Scholarship / Tuition Waiver for Graduate Studies — K.N. Toosi University of Technology
- 2013-06-01 Ranked 2nd in the Regional Skills Olympiad in Mechatronics (Qualified for the WorldSkills Competition, São Paulo, Brazil, 2015) — Esfahan Technical and Vocational Training Organization
- 2010-04-01 1st Place and Gold Coin Award at the Gravity Challenge — University of Payam Noor
- 2009-09-01 Top 0.4% in the Nationwide University Entrance Exam and Awarded a Full Scholarship / Tuition Waiver for Undergraduate Studies — Isfahan University of Technology

Standardised Scores

- Canadian English Language Proficiency Index Program: 9/12 (2023-08-01)

- International English Language Testing System: 7.5/9.0 (2021-01-01)
- Test of English as a Foreign Language: 98/120 (2017-10-01)

Engagement

Projects

- **AI-Powered Exam Preparation & Assessment Platform (Full-Stack Distributed SaaS Architecture)** 2026-04-01 – 2026-05-31
 - Architected an AI-powered examination & assessment platform using Next.js 15, React 18, TypeScript, Node.js, Express.js, PostgreSQL, Firebase Auth, Stripe, and Docker.
 - Designed a layered system architecture with SSR/CSR rendering, REST APIs, secure internal service communication, and containerized deployment via Docker Compose.
- **Fuzzy Control of Truck-Trailer Auto Parking in the Presence of Obstacles** 2013-10-01 – 2013-12-01
 - Developed a hierarchical fuzzy logic controller in MATLAB/Simulink for autonomous reverse parking of a truck-trailer system.
 - Modeled the nonlinear kinematics of a 4-DOF trailer-truck vehicle and implemented steering control algorithms for reverse maneuvering.
- **Design and Programming of a 3-DoF Robot Controller Circuit Board** 2013-04-01 – 2013-07-01
 - Designed and developed an embedded control system for a 3-DoF robotic manipulator using an ATmega32 AVR microcontroller.
 - Implemented simultaneous position and motion control of two servo motors and one DC motor to execute predefined robotic trajectories.
- **Control System Design of a Nonlinear Inverted Pendulum System** 2013-04-01 – 2013-06-01
 - Developed a mathematical model of a nonlinear inverted pendulum-on-cart system using Newtonian dynamics and derived governing equations of motion.
 - Linearized the nonlinear system around the upright equilibrium point and formulated both transfer function and state-space representations.
- **Hydraulic Circuit Design for Ring Rolling Machine** 2013-04-01 – 2013-06-01
 - Designed hydraulic circuits for a ring rolling machine based on specified force, stroke, pressure, and speed requirements.
 - Performed hydraulic actuator sizing calculations, including cylinder bore and piston rod diameter selection for multiple machine functions.
- **LabVIEW-Based Simultaneous Control of 3 CNC Stepper Motors** 2013-02-01 – 2013-03-01
 - Designed and developed a LabVIEW-based motion control system for simultaneous operation of 3 stepper motors used in CNC applications.
 - Implemented real-time motor speed control by converting user-defined rotational speed inputs into synchronized timing and pulse-delay sequences.
- **Stable Path Planning and Body Angle Analysis of a Biped Robot** 2012-03-01 – 2012-06-01
 - Analyzed the influence of torso/body angle parameters on the dynamic stability of a biped robot using Zero Moment Point (ZMP) stability criteria.
 - Developed kinematic and dynamic models of a multi-link biped robot, including body segments, joints, mass distribution, and inertia properties.

Volunteering

- **Table Tennis Umpire** — Huawei Technologies Canada 2025-04-01 – 2025-05-31
 - Corporate Table Tennis Tournament Officiating

- **Judge** — FIRST Robotics Canada 2023-03-01 – 2023-03-31
– FIRST Robotics Competition – University of Waterloo District Event
- **Technology Demonstrator** — University of Waterloo 2022-01-01 – 2024-04-30
– Robotics and Human-Machine Interaction Technology Demonstration
- **Mentor** — University of Waterloo 2019-12-01 – 2019-12-31
– ExpecTAtion Workshop for Teaching Assistants
- **Pit Administration Committee** — FIRST Robotics Canada 2019-03-01 – 2019-03-31
– FIRST Robotics Competition – Pit Administration Operations

Recommendations

- **Khaled Ahmed** (Colleague, Huawei Technologies Canada Co., Ltd.)
I know Ali from our time working in the same group. I had the opportunity to collaborate closely with him on several AI initiatives, particularly those focused on verifying and improving the reliability and robustness of large language model systems. Ali brings a rare combination of deep technical r...
- **Alireza Poostindouz** (Colleague, Huawei Technologies Canada Co., Ltd.)
Ali and I have known each other personally for over twenty years and professionally for more than four, and during our time working together at Huawei he consistently stood out as someone you can rely on. He's an exceptional AI and ML engineer with deep research experience and a strong publication r...
- **Frederic Risacher** (Manager, Huawei Technologies Canada Co., Ltd.)
I had the pleasure of hiring Ali into my team at Huawei to work on advancing autonomous driving safety. From the very beginning, I was impressed by his efficiency and technical expertise. Ali designed a monitoring-based system relying on safety metrics, leveraging his deep knowledge of state-of-the-...
- **Zohreh Aghababaeyan** (Colleague, Huawei Technologies Canada Co., Ltd.)
I am writing to recommend my colleague, Ali Nasr, for his work as a C++ programmer and algorithm designer at Huawei Inc. Ali contributed to the restructuring of the RTA monitor section's detailed design, integrating the RAMs team monitor functionality. His work was instrumental in providing key deta...
- **Clark R Dickerson** (Ph.D. Thesis Examiner, University of Waterloo)
I am delighted to write this letter of support for Ph.D. candidate Ali Nasr. Ali began his doctoral studies in September 2018, I reviewed his doctoral proposal, and provide collaboration opportunities, and consecutive advice over personal meetings. Since the start of the experimental data evaluation...
- **Maxime Raison** (Ph.D. Thesis Examiner, Ecole Polytechnique de Montreal)
I am delighted to write this letter of support for Ph.D. candidate Ali Nasr. I am external examiner of his PhD defense with the title of Design, Dynamics, and Control of Active-passive Upper-limb Exoskeleton Robots. Ali analyzed bioelectric signal and implemented a model that addressed the current c...
- **John Joseph McPhee** (Ph.D. Supervisor, University of Waterloo)
Outstanding progress, Ali. You continue to be the hardest-working member of our group, with an amazing record of papers published in the top-tier journals. Keep up the fantastic work!...
- **Hamid Reza Dokht Taghirad** (Teacher and M.Sc. Thesis Examiner, K. N. Toosi University of Technology)
I know Ali more than one years in the capacity of being his teacher for one graduate course and his thesis internal examiner. I find him hard working and an organaized student, resulting in being among top 5% student I have the pleasure to teach. In his term project research and in his master thesi...
- **Mehdi Keshmiri** (Undergraduate Supervisor, Isfahan University of Technology)
I have known Ali since September 2009 when he had started his education at Isfahan University of Technology (IUT). According to his high score in Robotics, Dynamic, Mechatronics, and Programming, along with his eagerness to do a high-quality project, I accepted to supervise his B.Sc. project. I have...